

Retail Sales EDA — Summary & Recommendations

Key findings from 1,000 transactions (Jan 2023 – Jan 2024)

Performance & Best Segments

The business generated **456,000 in revenue** (currency units) across **1,000 transactions** from 1,000 unique customers, averaging **456 per order** and **2,514 units sold**. Revenue is fairly balanced across categories, with **Electronics (156,905)** and **Clothing (155,580)** essentially tied as the top contributors and **Beauty (143,515)** trailing despite carrying the highest average price. **Adults (18-35)** are the strongest customer group - the largest segment (37%), the highest total spend (183,030), and the highest average order value (489) - while **Female customers** narrowly out-contribute Male customers (232,840 vs 223,160).

Trends & Weak Spots

Sales follow a clear seasonal pattern: they peak in **May** and again in **Q4** (Oct-Dec), and dip sharply through **Q3**, bottoming out in **September**. Weekly activity is highest on **Saturday** and lowest on **Thursday**. Purchase quantity shows almost no relationship to price (correlation of about 0.02), meaning demand is driven more by product type and timing than by cost. The clearest weak spots are the **Beauty category** (lowest revenue and volume), the **Q3/September slump**, and **Senior customers**, who have the lowest average order value of any group.

Recommendations

Management should: (1) build inventory and promotions around the May and Q4 peaks while running a targeted campaign to soften the September dip; (2) prioritize marketing spend on Adult customers, the highest-value segment; (3) revive Beauty through bundling or cross-promotion with Clothing and Electronics; and (4) use Saturday/Monday momentum for flash sales while testing Thursday-specific offers to smooth out the weekly demand curve.

Source: cleaned_dataset.csv and accompanying Power BI dashboard.